

Clean-Label Poisoning Attacks on Convolutional Neural Networks: From Feature Interpolation to Guided Diffusion Sampling

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Abstract—Clean-label data poisoning attacks manipulate a deep neural network’s training data without changing the ground-truth labels, which makes poisoned samples hard to catch under manual or automated inspection. Existing clean-label attacks, including feature collision, convex/bullseye polytope, and label-consistent adversarial perturbation methods, generally build poison samples through gradient-bounded pixel-space optimisation such as PGD. This approach is known to leave detectable high-frequency artefacts, and it is increasingly neutralised by diffusion-based purification defences such as DiffPure. This paper reports on a two-phase research programme that asks whether image generative models can produce clean-label poisons for convolutional neural network (CNN) image classifiers that are both more effective and harder to purify. Phase 1 develops a feature-interpolation poisoning method in which perturbations are derived from real target-class CNN embeddings via latent-space interpolation, then inverted to pixel space via PGD rather than initialised from random noise. Phase 2 replaces pixel-space inversion with guided reverse diffusion sampling from a pretrained denoising diffusion probabilistic model (DDPM), embedding a feature-collision objective directly into the denoising trajectory via SDEdit-style partial forward diffusion. We report CIFAR-10/ResNet-18 results for Phase 1 (Clean Data Accuracy 80.37%, Attack Success Rate 14.50%, mean SSIM 0.9993 at a 5% poison rate) and preliminary results for Phase 2, in which injecting only 25 diffusion-guided poison samples (approximately 0.05% of the training set) raises the target-class attack success rate from a 0.70% baseline to 30.80% while keeping the poisoned model’s clean accuracy at or above the clean-trained baseline. We further report a sweep over the partial-diffusion depth t_{start} , a preliminary cross-architecture (transfer-learning) evaluation, and a negative result showing the attack loses effect against a heavily ImageNet-pretrained victim. A purification-robustness comparison against DiffPure-style defences and against the Phase 1 PGD baseline is implemented but has not yet been run to completion, and is reported here as the primary remaining item of future work. We situate this work against the closest prior art, namely guided-diffusion poison generation and diffusion-based clean-label backdoor attacks on vision-language models, and lay out the specific, narrower novelty claim this research targets: single-stage, diffusion-native clean-label poison synthesis for CNN classifiers, to be benchmarked explicitly against purification-based defences.

Index Terms—clean-label poisoning, data poisoning, adversarial machine learning, diffusion models, convolutional neural networks, feature collision, adversarial purification

I. INTRODUCTION

A. Motivation

Machine learning systems increasingly depend on large, web-scraped, or crowd-sourced datasets, particularly in domains such as facial recognition, medical imaging, and autonomous driving. This dependence exposes models to *data poisoning* attacks, where an adversary injects manipulated samples into the training set to induce targeted misbehaviour at test time. *Clean-label* poisoning is a particularly stealthy variant of this: the attacker leaves ground-truth labels untouched, so poisoned samples pass both human visual inspection and automated label-consistency checks while still carrying a perturbation that biases the trained model [1].

The dominant family of clean-label attacks, feature collision [1], convex polytope [2], bullseye polytope [3], and label-consistent backdoors [4], builds poisons by adding an ℓ_p -bounded pixel perturbation to a base image, optimised via projected gradient descent (PGD) so that the poison’s feature representation collides with a target sample in a surrogate model’s embedding space. These methods work well, but they share two limitations that motivate this paper. First, the perturbation usually starts from random noise or an arbitrary trigger pattern rather than from semantically meaningful target-class structure, and this can hurt attack success rate (ASR) once end-to-end fine-tuning is applied across all model layers [1], [4]. Second, the resulting high-frequency, bounded-norm perturbations are increasingly well handled by diffusion-based purification and denoising defences, which have been shown to push the success of a broad range of existing clean-label attacks down to single digits with minimal loss of clean accuracy [8].

B. Research Overview

This paper reports on a two-phase research programme. The first phase established a clean-label poisoning baseline through a structured literature review, sorting existing clean-label backdoor attacks into three categories: feature collision (Poison Frogs [1], Hidden Trigger [5]), adversarial/generative perturbation methods (Label-Consistent [4], CleanSheet), and

surrogate-based inward-trigger methods (Narcissus). The review surfaced three gaps in the literature: trigger design that is arbitrary rather than semantically grounded, high attacker-knowledge requirements, and the largely unexplored use of generative models for poison image synthesis. In response we designed and implemented a *feature-interpolation poisoning* method (Phase 1), whose preliminary CIFAR-10 results are reported in Section V.

Building on the third gap, the second phase implements a diffusion-based poison generation pipeline that swaps out PGD-based pixel inversion for guided reverse diffusion sampling from a pretrained DDPM (Phase 2). The generation pipeline, surrogate-model training, and end-to-end victim-model poisoning/evaluation loop all run correctly, and Section V reports preliminary attack-success results, a sweep over the diffusion guidance depth, and a cross-architecture transfer evaluation. The purification-robustness comparison against DiffPure and against the Phase 1 PGD-based baseline (Section III) is implemented but has not yet been run to completion, and we report it here as ongoing work (Section VII).

C. Contributions

This paper makes the following contributions:

- A feature-interpolation clean-label poisoning method that derives perturbations from real target-class CNN feature prototypes rather than random initialisation (Section III, Phase 1).
- A single-stage guided-diffusion poisoning pipeline that embeds a feature-collision objective directly into the reverse denoising trajectory of a pretrained DDPM, requiring no diffusion model training (Section III, Phase 2).
- A positioning of this work against the closest prior art in diffusion-based poisoning, clarifying the specific methodological and evaluative gap it addresses (Section II).
- Preliminary quantitative results on CIFAR-10 with a ResNet-18 victim model for both phases, including a sweep over the diffusion guidance depth and a preliminary cross-architecture transfer evaluation, together with an evaluation protocol for the still-pending purification-robustness comparison (Section V).

II. RELATED WORK

A. Clean-Label Poisoning Attacks

Shafahi et al. [1] introduced feature collision poisoning, optimising a base image within an ℓ_∞ ball so its feature representation in a pretrained network collides with a target instance. This achieves high ASR (up to 100%) in transfer-learning settings but requires white-box access to the victim’s feature extractor, and it degrades substantially, to roughly 60–70% ASR, when the full network is retrained end-to-end. Zhu et al. [2] and Aghakhani et al. [3] improve transferability by surrounding the target with a convex or bullseye polytope of poison features rather than a single collision point. Turner et al. [4] combine adversarial perturbations with GAN-interpolated features to keep labels clean in an end-to-end

training setting, while surrogate-based inward-trigger methods such as Narcissus reach very high ASR (around 99%) at extremely low poisoning rates (around 0.05%) by training a surrogate on target-class and public out-of-distribution data alone. Two weaknesses show up consistently across this literature: reliance on white-box or otherwise substantial attacker knowledge, and vulnerability of the resulting bounded pixel perturbations to spectral and purification-based detection.

B. Diffusion Models for Poisoning and Purification

Diffusion models show up on both sides of this arms race. As a *defence*, DiffPure [6] and related diffusion-denoising methods forward-diffuse a possibly poisoned or adversarial input to an intermediate noise level and then denoise it, stripping out high-frequency perturbations while roughly preserving semantic content. This has been shown to push clean-label poisoning attack success close to zero across several classical attacks [8], and sparse diffusion purification methods such as ECLIPSE [7] target clean-label indiscriminate poisons directly, on the observation that poison perturbation morphology can be assimilated by progressive Gaussian noising. Separately, diffusion models have also been studied as a poisoning *target* in their own right, where a BadNets-style trigger is injected into a diffusion model’s own training data to study trigger memorisation and amplification [11]; this differs from our use of a diffusion model as a *poison-generation* tool for a downstream CNN classifier.

As an *attack* vector, Souri et al. (guided-diffusion poison generation) [9] use classifier-guided diffusion to synthesise clean-label *base images*, which are then handed off to an existing gradient-matching attack (e.g., Witches’ Brew) for the actual poisoning optimisation. This is a two-stage pipeline in which diffusion improves the starting point but does not itself carry out the feature-collision optimisation. More recently, CBV [10] modifies the score function during reverse diffusion generation to embed backdoor-trigger semantics directly into generated images, achieving over 80% ASR while preserving benign functionality. However, CBV targets clean-label backdoor attacks on vision-language models through multimodal (image and text) guidance, not feature-collision poisoning of CNN image classifiers.

C. Positioning of This Work

None of the work covered in this review performs *single-stage* guided reverse diffusion sampling, with a feature-collision objective embedded directly in the denoising trajectory and using only a frozen pretrained DDPM, for clean-label poisoning of CNN image classifiers. Nor does any of it benchmark the resulting poisons against diffusion-based purification defences relative to a PGD baseline. That is the specific, narrower gap Phase 2 of this paper targets. We are not claiming to be first to use diffusion models for poisoning in general; the claim is narrower: a specific combination of (a) single-stage guidance during sampling rather than a two-stage generate-then-attack pipeline, (b) application to CNN feature-collision poisoning rather than VLM backdoors, and (c) an

explicit purification-robustness evaluation against PGD-based baselines, which remains to be completed (Section VII).

III. RESEARCH DESIGN AND METHODOLOGY

A. Threat Model

Consistent with the clean-label poisoning literature, we assume an attacker who can inject a bounded number of correctly labelled poison samples into the victim’s training set but cannot modify labels, control the training procedure, or access the victim model’s final trained weights. The attacker trains a surrogate CNN (or, in Phase 2, uses a pretrained diffusion model together with a lightweight surrogate) to approximate the victim’s feature space, in line with standard black-box transfer assumptions in this literature [1], [3]. In Phase 2, the diffusion model acts purely as a poison generator and is neither the attack target nor the defence mechanism, which distinguishes this work from the diffusion-poisoning and diffusion-purification literature discussed in Section II.

B. Phase 1: Feature-Interpolation Poisoning

Phase 1 addresses the observation that prior methods initialise perturbations from random noise rather than semantically meaningful structure. The method has four stages, shown in Fig. 1:

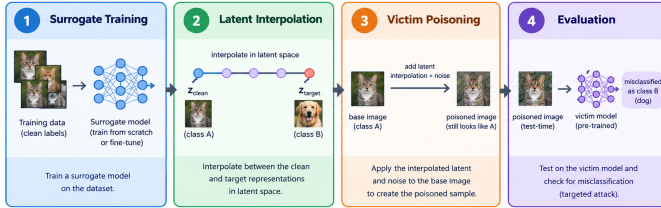


Fig. 1. Phase 1 methodology diagram showing the four-stage pipeline: surrogate training, latent interpolation, victim poisoning, and evaluation.

- 1) Surrogate training.** A ResNet-18 is fine-tuned on target-class images only, and the mean penultimate-layer embedding (prototype) p_t of the target class is extracted.
- 2) Latent interpolation.** A base image (from the class whose label the poison will retain) is encoded to a feature representation z_b . An interpolated target representation is computed as
$$\tilde{z} = (1 - \lambda)z_b + \lambda p_t, \quad \lambda \in [0, 1], \quad (1)$$
and inverted back to pixel space via PGD subject to an ℓ_∞ budget ϵ around the base image.
- 3) Victim poisoning.** The resulting poison images keep the base image’s ground-truth label and are injected into the training set; the victim model is fine-tuned on the union of clean and poisoned data.
- 4) Evaluation.** Clean Data Accuracy (CDA) and Attack Success Rate (ASR) are measured on held-out data, and

visual stealth is quantified via the Structural Similarity Index (SSIM) between base and poison images.

The interpolation coefficient λ gives a continuous, tunable trade-off between attack strength (higher λ , features closer to the target) and visual stealth (lower λ , appearance closer to the base image).

C. Phase 2: Guided Diffusion Sampling

Phase 2 replaces the PGD pixel-inversion step of Phase 1 with guided reverse diffusion sampling from a *pretrained, frozen* DDPM (google/ddpm-cifar10-32). The idea is that poisons generated natively through a diffusion process, rather than perturbations bolted onto a clean image via bounded pixel optimisation, should sit closer to the diffusion model’s own data manifold and so be harder for diffusion-based purification defences to strip away.

Algorithm 1 Guided Diffusion Poison Generation

Require: base image x_b , target image x_t , surrogate model $\phi(\cdot)$, partial-diffusion depth t_{start} , guidance scale g , LPIPS weight λ_{lp}

- $f_t \leftarrow \phi(x_t)$ ▷ target feature embedding
- Sample noise $\epsilon \sim \mathcal{N}(0, I)$
- $x_{t_{\text{start}}} \leftarrow \text{ForwardDiffuse}(x_b, \epsilon, t_{\text{start}})$ ▷ SDEdit
- for** $t = t_{\text{start}}$ **down to** 0 **do**
- $\hat{\epsilon} \leftarrow \text{UNet}(x_t, t)$
- $\hat{x}_0 \leftarrow \text{PredictClean}(x_t, \hat{\epsilon}, t)$
- $\mathcal{L} \leftarrow \|\phi(\hat{x}_0) - f_t\|_2^2 + \lambda_{\text{lp}} \cdot \text{LPIPS}(\hat{x}_0, x_b)$
- $g_t \leftarrow \nabla_{x_t} \mathcal{L}$
- $x_{t-1} \leftarrow \text{SchedulerStep}(\hat{\epsilon}, x_t) - g \cdot \frac{t}{T} \cdot g_t$
- end for**
- return** $x_{\text{poison}} = x_0$

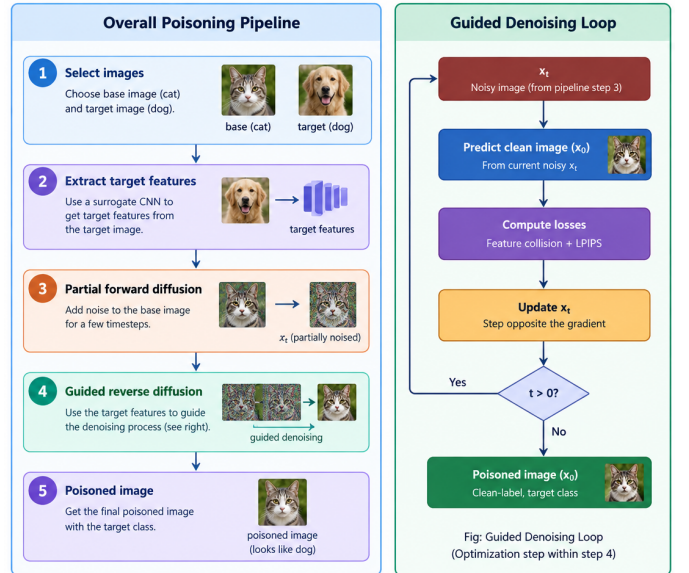


Fig. 2. Overall poisoning pipeline and guided denoising loop.

Algorithm 1 lays out the procedure. The base image is first partially forward-diffused to an intermediate timestep t_{start} (SDEdit-style), which keeps coarse semantic structure and the correct class label intact. During reverse denoising, at each step the model’s estimate of the clean image $\hat{x}_0$ is scored against a feature-collision loss (distance to the target’s surrogate-model embedding) and an LPIPS perceptual-distance penalty against the base image. The gradient of this combined loss then nudges the denoising trajectory, similar to classifier guidance in conditional diffusion sampling.

IV. ARTEFACT DEVELOPMENT

Both phases were implemented in Python using PyTorch and the HuggingFace `diffusers` library, and run on free-tier Google Colab/Kaggle GPU instances (NVIDIA T4), reflecting a resource-constrained research setting. No diffusion model was trained from scratch; Phase 2 reuses the publicly available pretrained `google/ddpm-cifar10-32` checkpoint. Key implementation components include:

- A lightweight ResNet-style surrogate CNN (four convolutional blocks, batch normalisation, global average pooling), trained from scratch on CIFAR-10 in roughly 2–3 minutes per run, used as the feature extractor for guidance in both phases.
- An LPIPS perceptual-distance module (AlexNet backbone) used as a stealth regulariser in both the Phase 1 interpolation-inversion step and the Phase 2 diffusion guidance loop.
- A `PoisonedCIFAR10` dataset wrapper that injects a configurable number of poison samples into the standard CIFAR-10 training split without altering existing clean samples or labels.
- A `DiffPure`-style purification module reusing the same pretrained DDPM used for poison generation, parameterised by a purification depth t_{purify} , for the planned robustness comparison.

The artefact runs the full Phase 1 pipeline and the full Phase 2 pipeline end to end, from poison generation through victim evaluation (data loading, surrogate training, guided poison generation, victim training, and single-target/aggregate ASR measurement), and this end-to-end loop has produced the preliminary results reported in Section V, including a sweep over the partial-diffusion depth t_{start} and a cross-architecture transfer run. As a qualitative sanity check, we additionally ran the Phase 2 pipeline on MNIST (using a separate pretrained DDPM, `laurent/ddpm-mnist`) and confirmed that a base digit’s diffusion trajectory can be guided towards a different target digit’s feature representation while remaining visually close to the base digit depending on the value of different hyperparameters (Fig. 3), indicating that the method is not specific to CIFAR-10.

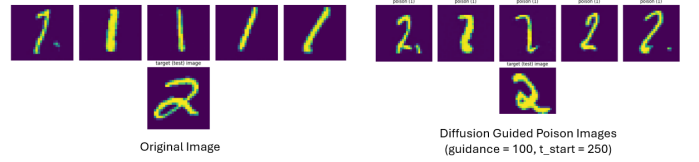


Fig. 3. MNIST proof-of-concept showing original digit images alongside their diffusion-guided poisoned counterparts.

V. EVALUATION

A. Phase 1 Preliminary Results

Table I reports a baseline run of the feature-interpolation method on CIFAR-10 with a ResNet-18 victim, using interpolation coefficient $\lambda = 0.35$, perturbation budget $\varepsilon = 0.0314$ (8/255), and a poisoning rate of 5% of the base class (250 samples).

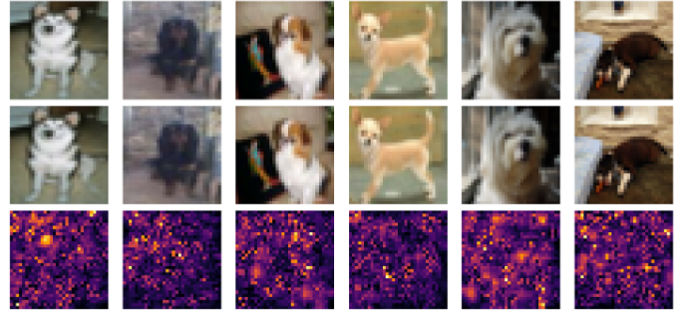


Fig. 4. Dog poison-pair visualization and amplitude-difference heatmaps. Rows 1–3 show the original images, poisoned images, and corresponding difference heatmaps, respectively.

TABLE I
PHASE 1 BASELINE RESULTS (CIFAR-10, RESNET-18)

Metric	Value
Clean Data Accuracy (CDA)	80.37%
Attack Success Rate (ASR)	14.50%
Mean SSIM (poison vs. base)	0.9993
ℓ_∞ budget violations	0 / 250

Table II places this preliminary result alongside reported figures from the literature. The current ASR (14.5%) sits well below reported figures for mature methods such as Narcissus (99.03%), which suggests that hyperparameter tuning, particularly of λ , ε , and the victim training schedule, is needed before the method can be considered competitive. This tuning is flagged as immediate future work (Section VII).

TABLE II
COMPARISON WITH LITERATURE (PRELIMINARY)

Method	Setting	Poison Rate	ASR	CDA	White-box?
Poison Frogs [1]	Transfer	1–5 samples	100%	~95%	Yes
Label-Consistent [4]	End-to-end	~1%	>50%	~95%	Yes
Narcissus	End-to-end	0.05%	99.03%	>99.6%	Partial
Ours (Phase 1)	End-to-end	5%	14.5%*	80.37%	Partial

*Early baseline result; optimisation of λ , ε , and training schedule needed.

B. Phase 2 Preliminary Results

The Phase 2 guided-diffusion pipeline (Algorithm 1) was evaluated on CIFAR-10 with target class “dog” and base class “cat”: the attack goal is to induce dog→cat misclassification by injecting 25 diffusion-guided poison samples (approximately 0.05% of the training set) with the base class’s own correct label. The surrogate CNN was trained for 15 epochs and used to extract the target-class feature signal for the diffusion guidance loop; both surrogate and victim CNNs were trained with the Adam optimiser, and the victim CNN was trained for 8 epochs, the setting used to compute CDA and ASR below.

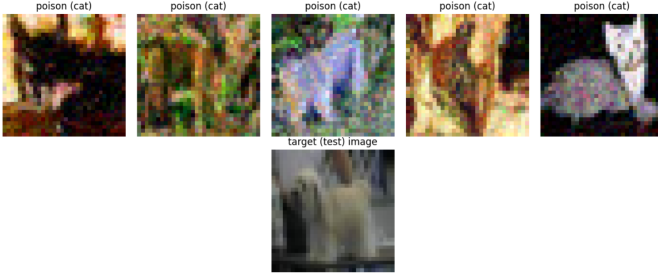


Fig. 5. Poison samples generated by diffusion, showing the CIFAR-10 class poison grid alongside the dog target image.

Table III reports the target-class ASR and clean accuracy as a function of the partial-diffusion depth t_{start} , at a fixed guidance scale of 100. “Clean-trained” and “Poison-trained” accuracy refer to the victim CNN’s overall test accuracy when trained on the unmodified and the poisoned training set, respectively; “Clean/Poisoned-model ASR” refer to the fraction of held-out dog images misclassified as cat by the corresponding model.

TABLE III
PHASE 2 RESULTS VS. PARTIAL-DIFFUSION DEPTH t_{start} (CIFAR-10, DOG→CAT, GUIDANCE SCALE = 100, 25 POISON SAMPLES)

t_{start}	Clean-trained acc.	Poison-trained acc.	Clean-model ASR	Poisoned-model ASR	ASR increase
50	71.45%	78.34%	0.80%	28.30%	+27.50 pp
100	72.89%	78.74%	0.70%	30.80%	+30.10 pp
300	71.45%	70.08%	0.80%	45.00%	+44.20 pp

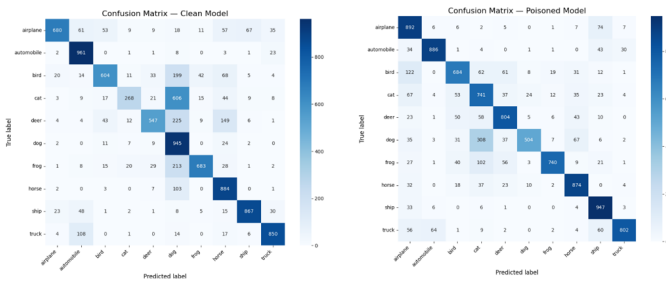


Fig. 6. Confusion matrices for the clean and poisoned models in the $t_{\text{start}} = 100$ run, showing the shift in the dog row from the dog column toward the cat column.

Across all three depths, injecting only 25 poisons (0.05% of the training set) shifts the dog→cat confusion rate far above the clean baseline. At $t_{\text{start}} = 100$, poisoned-model CDA (78.74%) is above the clean-trained baseline (72.89%), consistent with the clean-label constraint; at $t_{\text{start}} = 300$, however, the poisoned-model accuracy (70.08%) falls slightly below the clean baseline (71.45%) even as ASR rises to 45.00%, indicating that a deeper partial-diffusion step trades some of the clean-label stealth property for attack strength. This ASR/CDA trade-off as a function of t_{start} mirrors the λ trade-off in Phase 1 (Eq. 1) and is a natural target for the ablation study proposed in Section VII.

1) *Cross-Architecture (Transfer) Evaluation:* To test whether the attack transfers beyond the architecture used for guidance, we repeated the $t_{\text{start}} = 100$ setting with a victim architecture that differs structurally from the lightweight surrogate CNN: a compact ResNet-style network built from convolution-batchnorm-ReLU blocks with residual connections, global average pooling and a linear head, rather than the surrogate’s plain four-block convolutional stack. This is intended as a more meaningful cross-architecture test than simply widening or deepening the same architecture, since the residual connections introduce a genuine architectural difference while the two networks retain enough shared inductive bias to both be realistic CIFAR-10 classifiers.

TABLE IV
PHASE 2 CROSS-ARCHITECTURE (TRANSFER) RESULT (CIFAR-10, DOG→CAT, $t_{\text{start}} = 100$)

Metric	Value
Clean-trained victim accuracy	80.51%
Poison-trained victim accuracy	75.27%
Clean-model ASR	7.50%
Poisoned-model ASR	16.10%
ASR increase	+8.60 pp

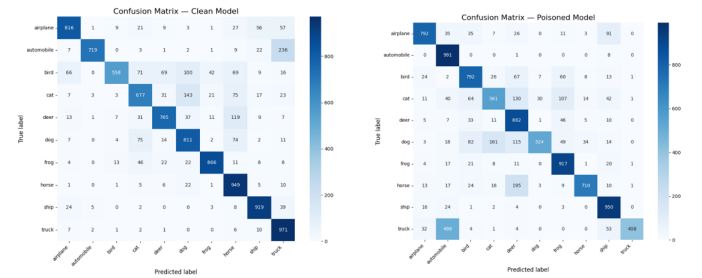


Fig. 7. Confusion matrices for the clean and poisoned models for the cross-architecture victim.

The attack still transfers to the structurally different victim, raising ASR by 8.60 percentage points, but the effect is markedly weaker than the same-architecture case in Table III (+30.10 pp), and poisoned-model accuracy drops visibly below the clean-trained baseline (75.27% vs. 80.51%). This suggests the current guidance signal is partly overfit to the surrogate architecture’s own feature geometry, consistent with the gen-

eral black-box transfer difficulty noted for clean-label attacks in Section II.

2) *A Negative Result: Strongly Pretrained Victims:* As a further robustness probe, we also attempted the attack against an ImageNet-pretrained ResNet fine-tuned on CIFAR-10 (with inputs resized from 32×32 to 224×224). In this setting the injected diffusion-guided poisons failed to shift the dog→cat decision boundary in any measurable way. We attribute this to the victim’s strong pretrained representation: a network initialised from large-scale, high-resolution ImageNet training already encodes robust class-relevant features, and the comparatively small amount of CIFAR-10 fine-tuning is unlikely to be sufficient to redirect that representation in the specific direction the clean-label attack requires. We report this as a genuine negative result and boundary condition on the threat model, rather than omitting it, since it indicates that the strength of the victim’s pretrained prior is itself a factor in clean-label poisoning susceptibility; we return to this point in Section VII.

C. Evaluation Protocol Going Forward

The full evaluation planned for the final version of this paper will additionally report:

- ASR and CDA as a function of poison budget (1%, 5%, 10%, 20% of the base class), extending the single-budget results above.
- Feature-collision distance degradation under DiffPure-style purification, compared between Phase 1 (PGD-based) and Phase 2 (diffusion-guided) poisons, across multiple purification depths t_{purify} .
- Additional target/base class pairs beyond dog/cat, to test the generality of the results in Section V-B.
- Robustness against standard backdoor/poisoning defences (Neural Cleanse, STRIP, Adversarial Neuron Pruning).
- Visual/statistical stealth via SSIM and LPIPS, to complement the ASR/CDA figures already reported.

VI. DISCUSSION

The preliminary Phase 1 results show that a feature-interpolation-based poison can be generated and injected into a CNN training pipeline while staying near-identical in appearance to the base image (SSIM = 0.9993) and without meaningfully degrading clean accuracy (80.37%), which is consistent with the clean-label constraint. The current ASR (14.5%) is a baseline figure rather than a final result and should improve with hyperparameter tuning. The lower white-box requirement compared with Poison Frogs (surrogate trained on target-class images only) goes some way toward addressing the attacker-knowledge gap identified in the literature review, though it does not remove the need for surrogate feature alignment with the victim.

The Phase 2 results in Section V-B indicate that single-stage guided diffusion sampling is a viable clean-label poisoning mechanism at an extremely low poisoning rate (0.05% of the training set): the same-architecture ASR increase (up to +44.20 pp at $t_{\text{start}} = 300$) is substantially larger than the current Phase

1 ASR at a $100\times$ higher poison rate, though the two are not directly comparable because they use different base/target class pairs and victim architectures. The t_{start} sweep shows a clear stealth/effectiveness trade-off analogous to the λ parameter in Phase 1: deeper partial diffusion increases ASR but can push poisoned-model accuracy below the clean baseline, which weakens the clean-label guarantee. The cross-architecture result shows the attack transfers, but with a substantially reduced effect and a larger clean-accuracy cost, and the negative result against an ImageNet-pretrained victim shows that the attack’s effectiveness is not architecture-invariant but depends on how strongly the victim’s representation is already anchored by prior training. Together these results scope the current claim more precisely than a single headline ASR number would: the method works reliably in the same-architecture, from-scratch setting, transfers partially to a structurally different from-scratch architecture, and does not yet transfer to strongly pretrained victims.

The central open question for Phase 2, whether diffusion-native poison generation is meaningfully more resistant to diffusion-based purification than PGD-based poisoning, remains unconfirmed empirically at the time of this submission. We flag this explicitly as the primary remaining contribution of the research programme rather than an established finding, and the paper’s claims are scoped accordingly.

VII. FUTURE WORK

The immediate next steps are:

- 1) **Phase 1 tuning.** Sweep λ , ϵ , and the victim training schedule to target ASR above 80% while maintaining CDA.
- 2) **Purification-robustness comparison.** Run the implemented purification-robustness comparison (Algorithm 1 poisons vs. PGD baseline, before/after DiffPure-style purification) to completion, at a statistically meaningful sample size.
- 3) **Cross-architecture and cross-defence evaluation.** Extend the preliminary cross-architecture result (Table IV) to additional victim architectures (e.g., VGG-16, DenseNet-121), and evaluate both phases against Neural Cleanse, STRIP, and Adversarial Neuron Pruning (ANP) defences.
- 4) **Understanding the pretrained-victim negative result.** Investigate why the attack fails against ImageNet-pretrained victims (Section V-B) — for example, whether a longer or higher-learning-rate fine-tuning schedule, or guidance computed from intermediate rather than final pretrained features, restores attack effectiveness.
- 5) **Ablation studies.** Quantify how much the interpolation coefficient λ / partial-diffusion depth t_{start} , guidance scale, LPIPS weight, poison rate, and surrogate training data size each contribute to attack effectiveness, building on the t_{start} sweep already reported in Table III.

Target venues for a full submission include IEEE S&P workshops and adversarial machine learning workshops co-located with major computer vision conferences.

VIII. CONCLUSION

This paper has presented the structure and preliminary results of a two-phase research programme on clean-label poisoning of CNN image classifiers, moving from a feature-interpolation poisoning method (Phase 1) to a single-stage guided-diffusion poisoning pipeline (Phase 2). We have positioned this work against the closest prior art in diffusion-based poisoning and purification, and clarified that the specific contribution being targeted, diffusion-native, single-stage clean-label poison synthesis for CNN classifiers evaluated explicitly against purification-based defences, has not been directly addressed by existing published work, even though it overlaps substantially in method with related two-stage and VLM-focused diffusion poisoning work. Phase 1 results are reported as an early baseline that needs further tuning. Phase 2 now has preliminary quantitative results: at a 0.05% poisoning rate, guided-diffusion poisons raise the target-class ASR from 0.70% to as much as 45.00% depending on the partial-diffusion depth, with a partial but weaker effect under cross-architecture transfer and no measurable effect against a strongly pretrained victim. The purification-robustness comparison central to this paper is still to be completed and will be reported in the final version of this paper.

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